UNO Q
CAMERA
TOF
AUDIO
SERVO BUS
POWER
IMU
PACK MONITOR
TOUCH AND FEET
UNO Q VIN BOOST
J1 JMEDIA 2x30 1.27 SMD male (GPEC212-3002A011C1AF equiv.)
J2 JMISC 2x30 1.27 SMD male (GPEC212-3002A011C1AF equiv.)
J3 TF31-22S-0.5SH 22p 0.5 FFC (IMX219)
U2 TCA9555PWR
U1 TCA9406DCU
J4 BM08B-SRSS-TB ToF
J8 SM02B-SRSS-TB electret mic
J7 S2B-PH-SM4-TB speaker
U4 PAM8302AASCR
U5 SN74LVC1G125DBVR
U6 SN74LVC1G125DBVR
Q1 MMBT3906
JP1 TXEN source: 1-2 AUTO, 2-3 GPIO PE7
J5 S3B-PH-SM4-TB servo bus
D1 BZT52C5V1S
D40 BAT54H,115
J6 S4B-PH-SM4-TB 2S pack in
Q2 AON6407 reverse-polarity P-FET
F1 SMD1812P300TF/16 3A PTC
U3 TPS62933DRLR
L1 FNR6045S6R8MT 6.8uH 3.3A
U7 LSM6DSV16X
J12 BM04B-SRSS-TB trunk IMU I2C
R22 5mR 2W shunt
U8 INA226AIDGSR
U9 TTP233H-BA6
J9 SM02B-SRSS-TB touch pad
J10 SM02B-SRSS-TB left foot contact
J11 SM02B-SRSS-TB right foot contact
C24 10pF C0G (DNP)
U10 TPS61088RHLR
L2 FXL0630-2R2-M 2.2uH
F2 JK-MSMD500L-12V 5A PTC
TP1 MCLK0
TP6 EXP_INT
TP4 BUS
TP2 V5
TP3 VBAT
TP5 GND
TP7 VIN12
H2 M3
H3 M3
H4 M3

PWR_FLAG: this rail has no power_out pin anywhere on it, so ERC
cannot see what drives it. The flag says a source is off-sheet.
A rail a regulator drives does NOT get one.

UNO Q head hat rev B. Board-to-board pinout and the camera block copied from Arduino's UNO Media Carrier ASX00083 so arduino-linux-config
camera0=type1-2lanes applies. 2S pack -> 5 mOhm shunt -> AON6407. PTC F2 -> TPS61088 11.9 V -> JMEDIA 57/59 VIN (UNO Q's own buck). PTC F1 -> TPS62933
5 V -> amplifier only; JMISC 54/56 open. Interrupts on 3.3 V MCU pins J2.3/7/9. D40 keeps the servo pull-ups off +3V3. One 3.3 V I2C bus (CCI0)
carries camera 0x10, expander 0x26, ToF 0x29, INA226 0x40, head IMU 0x6B and, on J12, the trunk IMU 0x6A.

this project / the robot
Sheet: /
File: unoq_hat.kicad_sch
Title: the robot UNO Q head hat
Size: A1 | Date: 2026-10-10 | **Rev: C**
KiCad E.D.A. 10.0.6 | Id: 1/1